Jaisidh Singh

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Google Scholar: https://scholar.google.com/citations?user=Z6fahScAAAAJ

Education

2024–present (expected 2026) **MSc in Machine Learning**, *Eberhard-Karls Universität Tübingen*, Tübingen.
GPA: 1.52/5.0 (German scale, 1.0 to 5.0 from best to worst)
Thesis (in association with OpenEuroLLM: Aaron Klein & Antonio Orvieto): scaling laws for hybrid LLMs

2020–2024 **B.Tech in AI & Data Science**, *Indian Institute of Technology*, Jodhpur.
Thesis: negation-aware compositional reasoning in vision-language models.

Selected Publications

2025 **(Almost) Free Modality Stitching of Foundation Models**, ***EMNLP 2025, ICLR-W 2025***,
Jaisidh Singh, Diganta Misra, Boris Knyazev, Antonio Orvieto, **(❝ 3 citations)** .

2025 **Learning the Power of "No": Foundation Models with Negations**, ***WACV 2025***,
*Jaisidh Singh**, Ishaan Shrivastava*, Mayank Vatsa, Richa Singh, Aparna Bharati, **(❝ 54 citations)** .

(Refer to Google Scholar for a complete list of publications, * indicates equal contribution.)

Work/Research Experience

Apr 2026 – present **Thesis Researcher with OpenEuroLLM & ELLIS Institute Tübingen**, *With Dr. Aaron Klein & Dr. Antonio Orvieto*.
- Researching loss-optimal allocation of dense attention in hybrid LLMs from the perspective of scaling laws for loss and hyper-parameters.
- Configuring high-performance distributed pretraining towards 1.5B scale models (multi-node ZeRO-2/FSDP sharding), informed by principled throughput profiling.

Oct 2025 – May 2026 **Software Engineer Graduate Assistant at Tübingen AI Center**, *With Prof. Peter Gehler*.
- Designed & deployed end-to-end real-time interactive computer vision systems for Stadtmuseum Tübingen. Profiled latency-quality tradeoffs across segmentations backbones to meet user constraints under museum UX requirements.
- Designed & delivered a hands-on workshop on LLM-agent orchestration with LangGraph at GCPR 2025, covering tool-use, memory, & MCP towards a scholarly deep-research multi-agent system.

Jan 2025 – present **Guest Researcher at MPI-IS Tübingen**, *With Dr. Antonio Orvieto*.
- Investigated grokking dynamics in transformers: showed CE+AdamW training does not admit kernel regime in transformers, and profiled Layer Normalization effects w.r.t. NTK spectral oscillation and accelerated grokking by up to $8\times$. Published: ICLR Trustworthy AI Workshop 2026.
- Designed & trained a hypernetwork enabling $10\times$ more FLOP-efficient modality stitching for VLMs such as CLIP and LLaVA (multimodal LLMs). Results verified across ViT-S/B/L, Sentence Transformers, & GPT-2/Qwen/Pythia backbones. Published: EMNLP 2025 Main Conference.

Summer 2022 & 2023 **Research Intern at Bosch Research India**, *With Dr. Amit A. Kale & Sonam Singh*.
- Designed an interpretable failure-discovery system for semantic segmentation models on industry-scale autonomous driving datasets. Automated detection of human-coherent segmentation errors via clustering methods achieving 95% precision. Technical report: arXiv:2411.10845 .
- Built multi-modal image retrieval pipelines using transformers for internal evaluation on weather-based image corruptions. Results informed efficacy of downstream profiling systems for autonomous driving data.

Summer 2022 & 2023 **Researcher at Trusted AI Lab IITJ**, *With Prof. Mayank Vatsa & Prof. Richa Singh*.
- Designed CC-Neg, a compositional negation benchmark & fine-tuned CLIP on a custom negation-augmented contrastive objective to achieved SOTA on negation understanding across diverse retrieval tasks. Published: WACV 2025 (50 citations).
- Systematically analysed StyleGAN2 latent spaces to profile identity leakage via dual-space identity invariance; results informed privacy-preserving generative model guidelines. Published: WACV 2024.

Technical Skills

- **Languages:** English (Fluent + Professional), Hindi (Native), German (A1)
- **Machine learning:** PyTorch, JAX/FLAX/Haiku, Huggingface
- **Distributed training & HPC:** Slurm, HTCondor, Data/Model/Tensor/Pipeline Parallelism
- **Agentic tools:** `dsPy`, LangGraph, LangChain, Claude Code, OpenCode
- **Programming languages:** Python, LaTeX, Bash, JavaScript, Dart, C++

Key Projects

`nano-dlm` **Diffusion Language model Pretraining Infrastructure**, *(jax, flax, optax)*, GitHub Link.
Clean, scalable, & end-to-end pretraining codebase for discrete diffusion LLMs. Features include multiple noise schedules (cosine, linear, square root), strong reproducibility via PRNG management, fully functional & config-driven training loop for side-effect free execution. Designed for extensibility towards large-scale research as well as hackable model speedrunning.

`loraclip` **LoRA fine-tuning library for CLIP**, *(PyTorch)*, GitHub Link (40 stars).
Minimal implementation (released as a `python` library) for rank-decomposed adaptation (LoRA) of CLIP's vision and text encoders; used by the community for parameter-efficient multimodal fine-tuning.

`pytorch-mixtures` **One-stop Solution for Mixture-of-Experts in LLMs**, *(PyTorch)*, GitHub Link (27 stars).
Minimalist & modular implementation of sparse mixture-of-experts such as TopK and ExpertChoice for transformer feed-forward layers; integrated with robust theory-driven testing of routing protocols.

Talks & Presentations

2026 **Explaining Grokking in Transformers from the Lens of Inductive Bias**, *ICLR-W 2026*.
Research Poster: https://www.dagm-gcpr.de/year/2025/program

2025 **Tutorial on AI Agents, LangGraph, & MCP**, *GCPR 2025*.
Workshop Article: https://www.dagm-gcpr.de/year/2025/program

2025 **(Almost) Free Modality Stitching of Foundation Models**, *Cohere Labs, EMNLP Main 2025*.
Research Presentation YouTube Video: https://youtu.be/tKvm9VYpqVw?si=h5VOrj9ZqFHMVAWD
Research Poster: https://jaisidhsingh.github.io/assets/pdf/HYMA_EMNLP_Poster-2.pdf

2025 **Tutorial on Continuous PDE Forecasting with INRs**, *University of Tübingen*.
Seminar Tutorial: https://jaisidhsingh.github.io/blog/2025/dinopinn/

2025 **Learning the Power of "No": Foundation Models with Negations**, *WACV 2025*.
Research Poster: https://jaisidhsingh.github.io/assets/pdf/wacv25-2841.pdf

2024 **SynthProv: Interpretable Framework for Profiling Identity Leakage**, *WACV 2024*.
Research Poster: https://jaisidhsingh.github.io/assets/pdf/synthprov_wacv2024_poster.pdf

Fellowships, Awards, & Achievements

- Konrad Zuse School (ELIZA) Fellowship for Master's students 2025-2026
- Konrad Zuse School (ELIZA) Fellowship for Master's students 2024-2025
- All-India-Rank 6428 in JEE Mains 2020, All-India-Rank 3214 in JEE Advanced 2020
- SOF awarded laptop for top regional ranks in Olympiads (2011)

Academic Service

- **Reviewer:** CVPR'26, ECCV'26, NeurIPs'26.
- **Teaching assistant:** Course: Understanding Language Models (Summer Semester '26 at University of Tübingen, with Prof. Carsten Eickhoff).